

OUTSTANDING RESEARCH AND TECHNICAL EFFORT

Not so very long ago it seemed that outstanding technical ability was vital only to a few highly scientifically oriented industries such as electronics, aerospace, pharmaceutical and chemical manufacturing. As these have grown, their ever-widening technologies have so penetrated virtually all lines of manufacturing and nearly all the service industries that today to have outstanding research and technical talent is nearly, if not quite, as important for a shoe manufacturer, a bank, a retailer or an insurance company as it is for what were once considered the exotic scientific industries that maintained large research staffs. Technological efforts are now channeled in two directions: to produce new and better products (in this connection, research scientists may, of course, do some-what more for a chemical company than for a grocery chain) and to perform services in a better way or at a lower cost than in the past. With regard to the latter objective, outstanding technical talent can be equally valuable for either group. Actually, in some of the service businesses, technological groups are opening up new product lines as well as paving the way for performing old services better. Banks are an example. Low-cost electronic input devices and minicomputers are enabling them to offer accounting and bookkeeping services to customers, thus creating a new product line for these institutions.

In research and technology, there is as much variation between the efficiency of one company and another as there is in marketing. In new product development, the complexity of the task almost guarantees this. Important as it is, the degree of technical competence or ingenuity of one company's research staff as compared to that of another is only one of the factors affecting the benefits that the company derives from its research efforts. Developing new products usually calls for the pooling of the efforts of a number of researchers, each skilled in a different technological specialty. How well these individuals work together (or can be induced by a leader to work together and stimulate each other) is often as important as the individual competence of the people involved. Furthermore, to maximize profits, it is vital not to develop just any product but one for which there will be significant customer demand, one that (nearly always)